Energy-Aware Adaptive Force-Admittance Control for Robotic Manipulators: Application in High-Precision Aircraft Assembly

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***Abstract*—An energy-aware adaptive force-admittance control strategy is proposed, which integrates accurate force regulation through a direct force controller with compliant motion enabled by adaptive admittance control. This combined mechanism addresses environmental uncertainties while simultaneously regulating wrench and position. However, the admittance adaptation and direct force controller introduce nonpassive behaviors, undermining the system's passivity. To resolve this issue, a virtual energy tank is introduced to monitor the energy associated with nonpassive actions, ensuring energy boundedness and restoring system passivity for stable interaction with passive environments. To further enhance safety and performance, the proposed scheme incorporates online energy injection and freezing mechanisms. When the tank's energy level falls below a preset threshold, additional energy is injected to improve control performance. Conversely, when the exerted wrench exceeds specified limits, the control system reverts to a constant admittance model to maintain compliance, effectively freezing any additional energy extraction. Two activation functions are also developed to ensure precise positioning in the direction of position control. The effectiveness of the proposed approach is validated through real industrial robot applications in aircraft assembly tasks. Experimental results demonstrate its superiority in improving both task performance and operational safety.**

***Index Terms*—Adaptive admittance control, aircraft assembly, energy tank, force control, passivity-based control.**

I. Introduction

PHYSICAL interaction between robots and complex, unstructured environments is indispensable for tasks such as material handling [1], precision machining [2], [3], and automated assembly [4]. However, these interactions often suffer from instability, particularly during tool engagement [5]. Hence, effective regulation of robot–environment interaction dynamics is paramount for reliable manipulation across diverse applications [6]. Existing interactive control strategies can be broadly categorized into direct force control, which offers precise wrench regulation; indirect schemes, notably impedance [7] and admittance [8] control, that enhance robustness in interaction dynamics; and passive compliance methods, which leverage intrinsic mechanical properties or passive dynamic elements to accommodate contact uncertainties. In [9], Lu et al. first proposed a novel inverse optimal adaptive controller for compliant robots, ensuring optimal tracking performance with significantly improved computation accuracy.

As robotic tasks continue to diversify, the need for integrated interaction controllers capable of simultaneously regulating motion and wrench has grown increasingly critical [10]. Aircraft assembly (see Fig. 1) serves as a representative example. Consequently, recent research efforts have focused on developing controllers that concurrently manage wrench and motion [11], [12], [13]. A prevalent solution is the hybrid force–position control framework of [14], which superimposes the force controller onto the motion controller. However, a principal drawback of hybrid force-position control is its inability to regulate wrench and motion along the same direction simultaneously [15]. To address this issue, the concept of the *selection matrix* was introduced, allowing wrench and motion to be decoupled into complementary subspaces [16]. However, this strategy depends on an accurate environmental model, which is often unavailable in practice [17].

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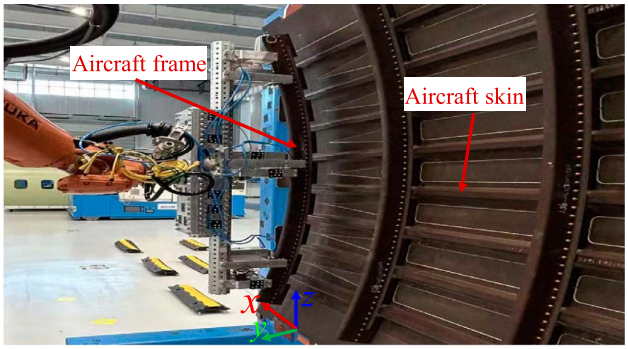


Fig. 1. Schematic diagram of the airplane panel assembly. During the assembly process, precisely aligning the frame and skin along the axial (x-axis) and circumferential (z-axis) directions is crucial for ensuring overall accuracy when joining different panels into sections. Meanwhile, proper control of the contact wrench in the normal (y-axis) direction is required to maintain adequate adhesion and avoid excessive stress, thereby preserving the structural integrity.

Adaptive force-impedance control methods have been developed to mitigate uncertainties in the environment. A pioneering study in [18] introduced a dynamic approach in which the selection matrix is continuously updated to accommodate newly imposed constraints. However, the inherent complexity and variability of the environment often render constraint estimation impractical. Moreover, frequent switching between controllers in response to changing constraints may compromise system stability [19]. To address these issues, parallel force-motion control methods have been proposed, which relax strict orthogonality requirements by prioritizing either force or motion control [20]. Nevertheless, improper prioritization can still lead to instability [15]. Other relevant studies that address environmental uncertainties include [21], [22], [23].

Estimating environmental constraints solely from wrench and motion measurements is often challenging. Accordingly, a force–motion control strategy capable of accommodating potential uncertainties is essential. This article proposes an integrated interactive controller that combines force and motion control, with a specific focus on aircraft assembly applications. In such scenarios, high-payload robots are indispensable for manipulating large-scale components. While industrial robots can handle heavy loads, their compliance is inherently limited under pure position control. To address this limitation, an admittance control framework is employed to adjust the robot's pose in response to external wrenches. However, admittance control is highly sensitive to model parameters [24], and parameter tuning typically demands substantial expertise. Furthermore, fixed stiffness and damping values may be insufficient for adapting to varying environmental conditions. To overcome these challenges, adaptive admittance control, which incorporates parameter adaptation into the admittance framework, has been proposed as a promising solution [25], [26]. This approach has proven effective in enhancing the compliance of industrial robots [27], [28].

Building on these findings, we propose an adaptive force–admittance control (AFAC) scheme. The framework employs direct force control to explicitly regulate the contact wrench and integrates adaptive admittance to enhance interaction performance in unmodeled environments. Stability is rigorously established via passivity theory, ensuring stable interaction with any passive environment.

The passivity property enables a comprehensive modeling of energy flow within the system, facilitating the identification of control actions that might compromise stability [29]. The concept of the virtual energy tank, introduced in [30], provides a pragmatic solution for preserving passivity by decoupling passive conditions from specific system dynamics [31]. Owing to its benefits, the energy tank approach has been widely adopted across various applications [32], [33], [34]. However, the issue of how to select an appropriate energy level has not been thoroughly investigated or adequately addressed in [26], [29], [30], [31], [32], [33], [34]. Notably, even if an interconnected system satisfies passivity through an energy tank, practical instability may still arise when the stored energy is excessive [35], while an insufficient energy level could hinder the system's ability to perform its intended task [36]. Therefore, a critical challenge for energy tank-based strategies is the effective management of the energy stored in the tank.

Various energy tank estimation methods have been proposed to address the challenge of selecting tank energy. In [37], the authors introduce the concept of task-energy, which estimates the energy required to complete a task. A similar approach is adopted in [15]. In [38], a strategy is proposed that initializes the energy tank at a high level and progressively lowers its minimum bound based on online task estimates. Additionally, the concept of energy freezing was introduced in [39], serving as an auxiliary energy storage layer that is released into the energy tank when necessary to complete tasks. However, these methods are not robust to uncertainties in robot–environment interactions. If the actual model parameters deviate from the nominal parameters used for task-energy estimation, the energy tank may still be depleted. This necessitates a real-time mechanism that determines, under predefined conditions, whether energy injection into the system is permissible. Specifically, if the system operates within a safe region (e.g., contact wrenches remain below a predefined threshold) but the tank lacks sufficient energy, real-time energy injection is employed to maintain performance. Conversely, once the contact wrench amplitude exceeds the threshold, the energy-extraction channel from the tank is immediately frozen to prevent further violations of passivity, thereby ensuring overall system safety.

Motivated by these considerations, this article develops an energy-aware AFAC for manipulators with applications in aircraft assembly. The key contributions are as follows.

1) An energy-aware AFAC model is proposed to enhance robustness against potential model dependencies, resulting in a stable controller that ensures wrench error convergence and/or precise positioning while maintaining safety and stability. In contrast to existing approaches [11], [13], [16], [17], [18], [23], the proposed method eliminates the traditional reliance on spatial reciprocity or environmental modeling.

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2) Unlike existing methods [10], [15], [26], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], which compute the tank energy offline or rely on estimated models, this article presents a model-independent energy tank that incorporates online energy injection and freezing mechanisms. The proposed energy tank dynamically injects or freezes energy, enabling an adaptive trade-off between performance and safety.
3) Two activation functions are introduced to ensure precise positioning during free-space while simultaneously preserving compliance during contact interactions. The framework is validated through real-world aircraft assembly experiments, encompassing both contact-free and contact-rich scenarios.

The article is organized as follows. Section II introduces the proposed AFAC model and discusses the stability challenges associated with the force controller and stiffness adaptation. Section III presents the energy-aware AFAC, while Section IV provides a detailed stability analysis. Experimental evaluations on an industrial robot, which serve to validate the theoretical framework, are described Section V. Finally, Section VI concludes the article.

Notations: The wrench is defined as $\boldsymbol{F} = [\boldsymbol{f}^\top, \boldsymbol{m}^\top]^\top \in \mathfrak{R}^6$, where $\boldsymbol{f} \in \mathfrak{R}^3$ denotes translational force, and $\boldsymbol{m} \in \mathfrak{R}^3$ represents rotational moment. Similarly, the pose of the robot's end-effector is expressed as $\boldsymbol{x} = [\boldsymbol{p}^\top, \boldsymbol{o}^\top]^\top \in \mathfrak{R}^6$, comprising the translational position $\boldsymbol{p} \in \mathfrak{R}^3$ and an appropriate representation of rotation, such as Euler angles $\boldsymbol{o} \in \mathfrak{R}^3$, defined in the Z–Y–X order. $\mathrm{diag}\{\}$ denotes a diagonal matrix, and $\emptyset$ signifies the empty set.

II. Problem Statement

To overcome the inherent limitation that industrial robots primarily support only position commands, this section presents an AFAC model, that facilitates the simultaneous regulation of both motion and wrench. Furthermore, the subsequent analysis rigorously examines the challenges associated with the stability of the proposed model.

A. AFAC Model

To achieve simultaneous regulation of motion and wrench, the AFAC model is formulated as follows:

$$\boldsymbol{M}\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}\tilde{\boldsymbol{x}} = \boldsymbol{F}_{\mathrm{err}} + \boldsymbol{F}_{\mathrm{ctrl}} \tag{1}$$

where $\boldsymbol{M} \in \mathfrak{R}^{6\times6}$, $\boldsymbol{B} \in \mathfrak{R}^{6\times6}$, and $\boldsymbol{K} \in \mathfrak{R}^{6\times6}$ are positive-definite diagonal matrices representing the constant mass, adaptive damping, and adaptive stiffness, respectively. The variables $\tilde{\boldsymbol{x}} = \boldsymbol{x}_r - \boldsymbol{x}_d \in \mathfrak{R}^6$, $\dot{\tilde{\boldsymbol{x}}} \in \mathfrak{R}^6$, and $\ddot{\tilde{\boldsymbol{x}}} \in \mathfrak{R}^6$ denote the deviations in the pose, velocity, and acceleration of the robot's end-effector. Here, $\boldsymbol{x}_d \in \mathfrak{R}^6$ represents the preplanned desired pose, while $\boldsymbol{x}_r \in \mathfrak{R}^6$ corresponds to the updated reference pose. The wrench tracking error is defined as $\boldsymbol{F}_{\mathrm{err}} = \boldsymbol{F}_e - \boldsymbol{F}_d$, where $\boldsymbol{F}_e \in \mathfrak{R}^6$ denotes the contact or external wrench, and $\boldsymbol{F}_d \in \mathfrak{R}^6$ represents the desired wrench. A direct force controller, $\boldsymbol{F}_{\mathrm{ctrl}}$, is implemented to mitigate the wrench tracking error, and is defined as follows:

$$\boldsymbol{F}_{\mathrm{ctrl}} = \boldsymbol{K}_p\boldsymbol{F}_{\mathrm{err}} + \boldsymbol{K}_i\boldsymbol{\rho}_{\mathrm{int}}(\boldsymbol{F}_{\mathrm{err}})\int \boldsymbol{F}_{\mathrm{err}}dt + \boldsymbol{K}_d\dot{\boldsymbol{F}}_{\mathrm{err}} \tag{2}$$

where $\boldsymbol{K}_p \in \mathfrak{R}^{6\times6}$, $\boldsymbol{K}_i \in \mathfrak{R}^{6\times6}$, and $\boldsymbol{K}_d \in \mathfrak{R}^{6\times6}$ are positive definite diagonal matrices representing the proportional, integral, and derivative gains of the direct force controller, respectively. The adaptive factor matrix is given by $\boldsymbol{\rho}_{\mathrm{int}}(\boldsymbol{F}_{\mathrm{err}}) = \mathrm{diag}\{\rho_{\mathrm{int},1}, \ldots, \rho_{\mathrm{int},6}\}$, where $\rho_{\mathrm{int},i}$, for $i \in \Theta_{\mathrm{total}}$, is defined as follows:

$$\rho_{\mathrm{int},i} = \sin^2\left(\frac{\pi}{2}\,\mathrm{clip}\left(\frac{|F_{\mathrm{err},i}| - \zeta}{\kappa_1|F_{d,i}| + \zeta}\right)\right) \tag{3}$$

where $\mathrm{clip}(*)$ is defined as $\mathrm{clip}(*) = \max(0, \min(*, 1))$. $\zeta > 0$ and $\kappa_1 \in (0, 1)$ are positive constants. The components $F_{\mathrm{err},i}$ and $F_{d,i}$ represent the ith elements of $\boldsymbol{F}_{\mathrm{err}}$ and $\boldsymbol{F}_d$, respectively. Consequently, $\rho_{\mathrm{int},i}$ is designed to smoothly suppress the integral gain, thereby mitigating overshoot when $|F_{\mathrm{err},i}|$ crosses $\kappa_1|F_{d,i}| + 2\zeta$. It decays to zero after the threshold $\kappa_1|F_{d,i}| + \zeta$, and remains at zeros when $|F_{\mathrm{err},i}| \le \zeta$.

B. Stability Analysis

To facilitate the stability analysis of system (1), a candidate *partial* Lyapunov function is defined as follows:

$$V_1 = \frac{1}{2}\dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{M}\dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\tilde{\boldsymbol{x}}^\top\boldsymbol{K}\tilde{\boldsymbol{x}} \tag{4}$$

which represents the potential and kinetic energy of the AFAC model. Taking the time derivative of V_1 yields

$$\dot{V}_1 = \dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{M}\ddot{\tilde{\boldsymbol{x}}} + \tilde{\boldsymbol{x}}^\top\boldsymbol{K}\dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\tilde{\boldsymbol{x}}^\top\dot{\boldsymbol{K}}\tilde{\boldsymbol{x}}. \tag{5}$$

Substituting the model in (1) into the above expression, we further obtain

$$\dot{V}_1 = \dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{F}_{\mathrm{err}} - \dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{F}_{\mathrm{ctrl}} + \frac{1}{2}\tilde{\boldsymbol{x}}^\top\dot{\boldsymbol{K}}\tilde{\boldsymbol{x}}. \tag{6}$$

The signs of $\dot{\tilde{\boldsymbol{x}}}^\top\boldsymbol{F}_{\mathrm{ctrl}}$ and $(1/2)\tilde{\boldsymbol{x}}^\top\dot{\boldsymbol{K}}\tilde{\boldsymbol{x}}$ are a priori unknown, which poses significant challenges in ensuring passivity with respect to the port $\langle\Delta\dot{\tilde{\boldsymbol{x}}}, \boldsymbol{F}_{\mathrm{err}}\rangle$. Moreover, although the AFAC model in (1) is generally applicable across all operational states, including free motion and contact phases, the compliance required during contact often comes at the expense of motion accuracy in free space. Consequently, further refinement of the control strategy is necessary to simultaneously ensure passivity and maintain performance. To support controller development, the complete set of dimensions is defined as $\Theta_{\mathrm{total}} = \{1, \ldots, 6\}$, while $\Theta_f \subseteq \Theta_{\mathrm{total}}$ denotes the set of force control directions.

Control objective: This article focuses on designing an energy-aware adaptive force-admittance controller for industrial robotic systems, with the aim of ensuring the following.

1) Guarantee the stability of the interconnected system comprising the AFAC system and the energy tank.
2) Ensure the convergence of wrench tracking errors $F_{\mathrm{err},i}, \forall i \in \Theta_f$, under the condition

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$\mu - \varepsilon_i = 1, \forall i \in \Theta_{\text{total}}$, where μ and ε_i are defined in (10) and (11), respectively.

3) Achieve predetermined positioning accuracy in free space when all directions are position-controlled.

The subsequent section discusses enhancements to the control strategy aimed at mitigating potential nonpassivity issues arising from the force controller and stiffness adaptation.

III. Design of an Energy-Aware AFAC for Enhanced Robotic Performance

In this section, a systematic design process for an energy-aware AFAC tailored for industrial robot systems is presented. First, two activation functions are introduced to enable smooth transitions between contact-free and contact-rich scenarios. Subsequently, an energy-aware AFAC model is formulated to achieve an adaptive balance between safety and performance.

A. Development of Activation Functions for Force and Admittance Controller

To ensure tracking performance in free space, the activation of PID force controllers during free-space motion must be avoided. Similar to (3), a PID force controller activation function matrix, denoted as $\boldsymbol{\rho}_{\text{frc}} = \text{diag}\{\rho_{\text{frc},1}, \ldots, \rho_{\text{frc},6}\}$, is proposed, where each element $\rho_{\text{frc},i}$, $\forall i \in \Theta_{\text{total}}$, is defined as follows:

$$\rho_{\text{frc},i} = \sin^2\left(\frac{\pi}{2}\,\text{clip}\left(\frac{\Delta F_{e,i}}{\delta_f}\right)\right). \tag{7}$$

Here, $\Delta F_{e,i} = |F_{e,i}| - \kappa_{3,i}$, with $\kappa_{3,i} > 0$ representing a positive constant, and $\delta_f > 0$ denoting the width of the activation region. The value of $\kappa_{3,i}$ is selected to exceed the sensor drift or measurement noise, thereby preventing unintended activation of the adaptive force-admittance controller in free space.

Similarly, to ensure both free-space tracking performance and compliance during environmental interaction, an adaptive admittance controller activation function matrix, $\boldsymbol{\rho}_f = \text{diag}\{\rho_{f,1}, \cdots, \rho_{f,6}\}$, is defined. Although $\rho_{f,i}, \forall i \in \Theta_{\text{total}}$, can be directly set as $\rho_{f,i} = \rho_{\text{frc},i}$, the adaptive admittance controller may not activate properly if the desired contact point $x_{d,i}$ planned along the force control direction, $\forall i \in \Theta_f$, has not yet engaged with the environment. In such cases, even if the actual robot pose x_i reaches $x_{d,i}$, the condition $|F_{e,i}| \le \kappa_{3,i}$ may still hold, resulting in $\rho_{f,i} = 0$. To address this limitation, an additional condition is introduced: once the positioning error $\|e_{x,i}\| = \|x_{d,i} - x_i\|$ falls below a predefined threshold $\kappa_4 > 0$ (i.e., $||e_{x,i}|| < \kappa_4$), the activation function $\rho_{f,i}$ is triggered for $i \in \Theta_f$. This ensures that, even if the measured contact wrench remains below the threshold due to contact point detection uncertainties or sensor noise, the force-admittance controller is still activated as the end-effector approaches the planned contact point location. As a result, the contact wrench can converge smoothly to the desired value. The formulation of $\rho_{f,i}$, $\forall i \in \Theta_{\text{total}}$, is given by

$$\rho_{f,i} = \max\left\{\sin^2\left(\frac{\pi}{2}\,\text{clip}\left(\frac{\Delta F_{e,i}}{\delta_f}\right)\right), \alpha_i \cdot \cos^2\left(\frac{\pi}{2}\,\text{clip}\left(\frac{|e_{x,i}|}{\kappa_4}\right)\right)\right\} \tag{8}$$

where $\alpha_i = 1$ if $i \in \Theta_f$ and $\alpha_i = 0$ otherwise.

Based on the activation functions defined in (7) and (8), the AFAC model in (1) is reformulated as follows:

$$\boldsymbol{M}\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}\tilde{\boldsymbol{x}} = \boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \boldsymbol{\rho}_{\text{frc}} \boldsymbol{F}_{\text{ctrl}}. \tag{9}$$

Remark 1: As previously noted, traditional hybrid force–position controllers may encounter conflicts between desired motion control and wrench regulation. To resolve this issue, adaptive admittance control is employed to ensure contact compliance; however, the positioning accuracy during free motion is often compromised. To address this limitation, two activation functions, as defined in (7) and (8), are proposed to ensure precise positioning in contact-free scenarios while maintaining compliance in contact-rich situations. Since the control mode must dynamically switch between precise tracking and active compliance, it is essential that the transition be smooth to avoid abrupt changes in control input. In contrast to the threshold-based switching strategies adopted in [40], the proposed activation functions utilize $\sin^2(\cdot)$ and $\cos^2(\cdot)$ to ensure smooth transitions. This design helps avoid abrupt binary switching, thereby reducing the risk of chattering in the presence of sensor noise or near-threshold fluctuations.

In the subsequent section, an energy tank equipped with online energy injection and freezing mechanisms is proposed to ensure the system passivity.

B. Energy-Aware AFAC Design

To address the passivity violation issue, a virtual tank is introduced for potential active behaviors in (9), namely, $\dot{\tilde{\boldsymbol{x}}}^{\top} \boldsymbol{\rho}_{\text{frc}} \boldsymbol{F}_{\text{ctrl}}$ and $(1/2)\tilde{\boldsymbol{x}}^{\top}\dot{\boldsymbol{K}}\tilde{\boldsymbol{x}}$. The energy in the tank is defined as $V_\vartheta = 0.5\vartheta^2$. Since the energy tank dynamics exhibit singularity when $\vartheta = 0$, a factor μ is introduced to ensure that the energy level of the tank remains above the predefined lower bound $V_{l,\vartheta} > 0$. It is defined as follows:

$$\mu = \begin{cases} 1, & V_\vartheta \ge V_{l,\vartheta} + \delta_{ek} \\ \sin^2\left(\frac{\pi}{2} \cdot \text{clip}\left(\frac{V_\vartheta - V_{l,\vartheta}}{\delta_{ek}}\right)\right) & V_{l,\vartheta} \le \vartheta \le V_{l,\vartheta} + \delta_{ek} \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where $\delta_{\text{ek}} > 0$ is a threshold. In existing tank-based methods, energy extraction by the system is prohibited when $\mu = 0$. Thus, determining the energy stored in the tank is critical for evaluating the system's ability to meet task requirements and adapt to uncertain environments [15]. However, uncertainties in the robot and environment render the offline determination of this value challenging.

To address this challenge, we propose a dynamic energy control strategy that actively regulates the energy stored in the tank during operation. The energy injection and freezing matrix is defined as $\boldsymbol{\varepsilon} = \text{diag}\{\varepsilon_1, \ldots, \varepsilon_6\}$, where each function $\varepsilon_i, \forall i \in \Theta_{\text{total}}$, governs online energy injection and freezing along the ith dimension. This mechanism facilitates controlled energy exchange between the robotic system and the tank under predefined conditions. Specifically, when the tank's energy is

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insufficient and the wrench magnitude falls below a threshold $F_{\mathrm{up},i} \geq |F_{\mathrm{d},i}|$ (i.e., $F_{e,\mathrm{sa},i} = |F_{e,i}| - F_{\mathrm{up},i} < 0$ and $\mu < 1$), leading to performance degradation, online energy injection is activated to restore system performance (i.e., $\mu - \varepsilon_i = 1$). Conversely, when the wrench magnitude becomes excessive (i.e., $F_{e,\mathrm{sa},i} \geq 0$), energy extraction from the tank is halted (i.e., $\mu - \varepsilon_i = 0$) to ensure system safety. While ε_i could be defined as a binary signal, a smooth transition between energy injection and freezing is ensured by defining it as follows:

$$\varepsilon_i = \mu - \phi_i \tag{11}$$

where

$$\phi_i = \cos^2\left(\mathrm{clip}\left(\frac{|F_{e,i}| - (F_{\mathrm{up},i} - \eta_{4,i})}{\eta_{4,i}}\right) \cdot \frac{\pi}{2}\right) \tag{12}$$

with $\eta_{4,i} > 0$ representing a threshold.

Consequently, the energy-aware AFAC model and the associated tank state, denoted by ϑ, are formulated as follows:

$$\boldsymbol{M}\Delta\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}\Delta\tilde{\boldsymbol{x}} = \boldsymbol{\rho}_f \boldsymbol{F}_{\mathrm{err}} + (\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon})\,\boldsymbol{\rho}_{\mathrm{frc}}\boldsymbol{\psi}_1 \boldsymbol{F}_{\mathrm{ctrl}} \tag{13}$$

$$\dot{\vartheta} = \frac{\varpi}{\vartheta}\left(\nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} - E\right) \tag{14}$$

where $0 \leq \nu < 1$ is a constant. Here, $\Delta\dot{\tilde{\boldsymbol{x}}} = \dot{\boldsymbol{x}}_r - \Delta\dot{\boldsymbol{x}}_d$ represents the velocity adjustment based on the energy level correction, and the ith component of the corrected planned velocity vector $\Delta\dot{\boldsymbol{x}}_d$ is given by $\Delta\dot{x}_{d,i} = (\mu - \varepsilon_i)\,(\dot{x}_{d,i} + x_{d,i} - \Delta x_{d,i})$ with $\Delta x_{d,i}(t_0) = x_{d,i}(t_0)$, where t_0 represents the initial time. Subsequently, $\Delta\boldsymbol{x}_d$ and $\Delta\ddot{\boldsymbol{x}}_d$ can be obtained by integrating and differentiating $\Delta\dot{\boldsymbol{x}}_d$, respectively.

The stored tank energy is exploited to restore passivity for both the stiffness adaptation and the PID force controller by defining E in (14) as follows:

$$E = \Delta\dot{\tilde{\boldsymbol{x}}}^{\top}(\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon})\left(\boldsymbol{\rho}_f\boldsymbol{F}_{\mathrm{err}} + \boldsymbol{\psi}_1\boldsymbol{\rho}_{\mathrm{frc}}\boldsymbol{F}_{\mathrm{ctrl}}\right) + \frac{1}{2}\Delta\tilde{\boldsymbol{x}}^{\top}\dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \tag{15}$$

where the update law $\dot{\boldsymbol{K}} = \mathrm{diag}\{\dot{K}_1, \ldots, \dot{K}_6\}$ is expressed as follows:

$$\dot{K}_i = \mathrm{Proj}\left((\mu - \varepsilon_i)\,\varphi_{2,i}\Pi_i, K_{\min,i}, K_{\max,i}\right), i = 1, \ldots, 6 \tag{16}$$

with $\Pi_i = -\rho_{f,i}\eta_1|\Delta\tilde{x}_i|F_{\mathrm{err},i}$ and $\eta_1 > 0$ representing the adaptation gain. Additionally, $\mathrm{Proj}(\cdot)$ denotes the projection operator [41]. This operator ensures that K_i remains confined within the predefined stiffness range $[K_{\min,i}, K_{\max,i}]$, assuming that the initial value of K_i lies within this interval. Here, $K_{\min,i}$ and $K_{\max,i}$ denote the minimum and maximum achievable stiffness values, respectively. Furthermore, limiting the energy extraction rate via a power constraint enhances system safety [42]. The term $\varphi_{2,i}, \forall i \in \Theta_{\mathrm{total}}$, is defined as an adaptive power scaling coefficient for stiffness adaptation. Additionally, we define $\boldsymbol{\psi}_1 = \mathrm{diag}\{\varphi_{1,1}, \ldots, \varphi_{1,6}\}$, where each $\varphi_{1,i}, \forall i \in \Theta_{\mathrm{total}}$, serves as an adaptive power scaling coefficient for the PID force controller. The specific form of $\varphi_{j,i}$, for $j \in \{1, 2\}$ and $i \in \Theta_{\mathrm{total}}$, is given by

$$\varphi_{j,i} = \begin{cases} \frac{p^+_{j,i}}{p_{j,i}}, & \text{if } p_{j,i} > p^+_{j,i} \\ 1, & \text{otherwise} \end{cases} \tag{17}$$

where $p_{2,i} = \mathrm{Proj}\left((\mu - \varepsilon_i)\,\Pi_i, K_{\min,i}, K_{\max,i}\right)$, and $p_{1,i} = (\mu - \varepsilon_i)\,\Delta\dot{\tilde{x}}_i\rho_{\mathrm{frc},i}F_{\mathrm{ctrl},i}$. To dynamically optimize the maximum power injection rate $p^+_{j,i}$, the following update law is proposed:

$$\dot{p}^+_{j,i} = (1 - \varphi_{j,i})\beta\,|F_{\mathrm{err},i}| - \lambda p^+_{j,i} \tag{18}$$

where $\beta > 0$ represents the learning rate, and $\lambda > 0$ is a forgetting factor to prevent excessively high injection rates.

In (14), the parameter ϖ ensures that the stored energy value, V_ϑ, in the energy tank does not exceed the predefined upper bound $V_{u,\vartheta}$. Accordingly, ϖ is defined as follows:

$$\varpi = \begin{cases} 0, & \text{if } V_\vartheta \geq V_{u,\vartheta} \text{ and } \nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq E \\ 1, & \text{otherwise.} \end{cases} \tag{19}$$

From (19), it is evident that when $\varpi = 1$, the inequality $(1 - \varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq (1 - \varpi)E$ holds trivially. For the case $\varpi = 0$, by definition, we have $\nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq E$, which again implies $(1 - \varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq (1 - \varpi)E$. Thus, the following inequality always holds:

$$(1 - \varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^{\top}\boldsymbol{\rho}_f\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq (1 - \varpi)E. \tag{20}$$

Remark 2: Fig. 2 illustrates the three operational modes of the energy tank. When $|F_{e,i}| \leq \left(F_{up,i} - \eta_{4,i}\right)$ and $\mu = 1$, the tank energy is sufficient and the system remains within the safe region; hence, no energy injection or freezing is required, as $\varepsilon_i = 0$ and $\mu - \varepsilon_i = 1$. When $|F_{e,i}| \leq \left(F_{up,i} - \eta_{4,i}\right)$ and $\mu < 1$, it follows that $\varepsilon_i = \mu - 1 < 0$. Under these conditions, energy is injected into the system in real time to maintain performance, ensuring that $\mu - \varepsilon_i = 1$. Conversely, when $|F_{e,i}| \geq F_{up,i}$, one obtains $\varepsilon_i = \mu$. In this case, the energy extraction from the tank is temporarily frozen by setting $\mu - \varepsilon_i = 0$. In this state, the AFAC degrades to constant admittance control (CAC), thereby ensuring the system's stability and safety. It is worth noting that the conditions for energy injection or freezing can be defined based on various control objectives or scenarios, such as maximum kinetic energy, contact/external wrench, or maximum speed limitations. In this article, to guarantee the convergence of the wrench error in the force control direction, the contact wrench/external wrench is employed as the condition for online energy injection or freezing.

The stability and performance of the developed method are analyzed in the following section.

IV. Stability and Performance Analysis

In this section, we first establish the stability of the proposed method. Subsequently, under the condition of $\mu - \varepsilon_i = 1$, the wrench tracking performance along the force control direction is examined. Finally, the system's precise positioning capability in free space is analyzed.

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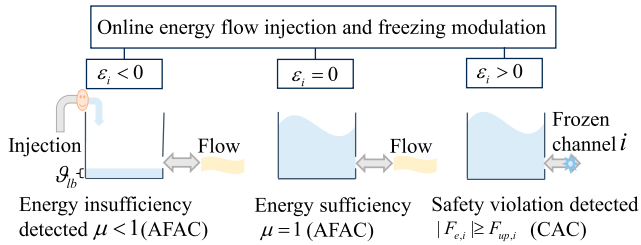


Fig. 2. Illustration of the energy injection and freezing concepts.

Let $\Theta = \{i \mid \varepsilon_i = \mu - \phi_i \text{ with } \phi_i < 1, \forall\, i \in \Theta_{\text{total}}\}$. To analyze system stability and the convergence of the wrench tracking error $F_{\text{err},i}$ for all $i \in \Theta_f$ under the condition $\Theta = \emptyset$, the total system state is defined as follows:

$$\boldsymbol{\chi} = \left[\Delta\tilde{\boldsymbol{x}}^\top, \Delta\dot{\tilde{\boldsymbol{x}}}^\top, \vartheta, \boldsymbol{x}_{\text{env}}^\top\right]^\top \tag{21}$$

where $\boldsymbol{x}_{\text{env}}$ denotes the environmental state corresponding to the environmental energy $V_{\text{env}}(\boldsymbol{x}_{\text{env}})$. Given a desired equilibrium level ϑ^* and $\boldsymbol{x}^*_{\text{env}}$, the set of all possible values of $\boldsymbol{\chi}$ is represented by Ω. Then, one has

$$\forall \boldsymbol{\chi} \in \Omega \backslash \left\{\left[\boldsymbol{0}^\top, \boldsymbol{0}^\top, \vartheta^*, \boldsymbol{x}^{*T}_{\text{env}}\right]^\top\right\}, V_2 > V_\vartheta(\vartheta^*) + V_{\text{env}}(\boldsymbol{x}^*_{\text{env}}) \tag{22}$$

where V_2 represents the total energy of the closed-loop system, comprising the AFAC dynamics, the energy tank, and the passive environment, and is defined as follows:

$$V_2 = \frac{1}{2}\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{M}\Delta\dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \boldsymbol{K}\Delta\tilde{\boldsymbol{x}} + V_\vartheta(\vartheta) + V_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right). \tag{23}$$

To establish the stability of the energy-aware AFAC scheme, the following assumptions are made:

Assumption 1: [15] The environment with which the robot interacts is passive.

Assumption 2: The reference trajectories $\boldsymbol{x}_d$ and the desired wrenches $\boldsymbol{F}_d$ are bounded for all $t \geq 0$.

Since the passive environment is inherently dissipative in nature [43], it follows that $\dot{V}_{\text{env}} < 0$ whenever $\boldsymbol{x}_{\text{env}} \neq \boldsymbol{x}^*_{\text{env}}$. The main results of this article can be summarized as follows.

Theorem 1: The energy-aware AFAC model described in (13) is demonstrated to be passive with respect to $\langle \Delta\dot{\tilde{\boldsymbol{x}}}, \boldsymbol{F}_{\text{err}}\rangle$ when $\Theta \neq \emptyset$ and asymptotically stable at the equilibrium point $\boldsymbol{\chi}^* = \left[\boldsymbol{0}^\top, \boldsymbol{0}^\top, \vartheta^*, \boldsymbol{x}^{*T}_{\text{env}}\right]^\top$ when $\Theta = \emptyset$. This stability is guaranteed provided that the energy tank dynamics $\dot{\vartheta}$, and the update laws $\dot{\boldsymbol{K}}$ are designed in accordance with (14), and (16), respectively. At the equilibrium point $\boldsymbol{\chi}^*$, $F_{e,i}$ converges to $F_{d,i}$ for all $i \in \Theta_f$ when $\Theta = \emptyset$. Furthermore, if all directions are position-controlled, the robot achieves a predetermined positioning accuracy ς in free space, such that $||\boldsymbol{x} - \boldsymbol{x}_d|| \leq \varsigma$.

Proof: Combining (15) and (20), we have

$$\begin{aligned}&(1-\varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \geq \\ &(1-\varpi)\left(\Delta\dot{\tilde{\boldsymbol{x}}}^\top (\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon})\left(\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \boldsymbol{\psi}_1 \boldsymbol{\rho}_{\text{frc}} \boldsymbol{F}_{\text{ctrl}}\right)\right. \\ &\left. + \frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}}\right).\end{aligned} \tag{24}$$

When $\rho_{f,i} > 0$ or $\rho_{\text{frc},i} > 0, \forall i \in \Theta_{\text{total}}$, the system stability is analyzed by defining the total energy of the closed-loop system with respect to the equilibrium point $\boldsymbol{\chi}^*$ as follows:

$$V(\boldsymbol{\chi}) = V_2 - V_\vartheta(\vartheta^*) - V_{\text{env}}\left(\boldsymbol{x}^*_{\text{env}}\right) \tag{25}$$

where V_2 is specified in (23). It can be shown that $V(\boldsymbol{\chi}^*) = 0$. Moreover, based on (22), it follows that $\forall \boldsymbol{\chi} \backslash \boldsymbol{\chi}^*, V(\boldsymbol{\chi}) > 0$. Thus, $V(\boldsymbol{\chi})$ qualifies as a valid Lyapunov function. The time derivative of the Lyapunov function V is expressed as follows:

$$\begin{aligned}\dot{V} = &\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{M}\Delta\ddot{\tilde{\boldsymbol{x}}} + \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{K}\Delta\tilde{\boldsymbol{x}} \\ &+ \frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\tilde{\boldsymbol{x}} + \vartheta\dot{\vartheta} + \dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right).\end{aligned} \tag{26}$$

By substituting the energy-aware AFAC model and the dynamics of the energy tank from (13) and (14), and after simplification, the expression becomes

$$\begin{aligned}\dot{V} = &\Delta\dot{\tilde{\boldsymbol{x}}}^\top \left(\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + (\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon})\,\boldsymbol{\rho}_{\text{frc}}\boldsymbol{\psi}_1 \boldsymbol{F}_{\text{ctrl}}\right) - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ \frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} + \varpi(\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} - E) + \dot{V}_{\text{env}}(\boldsymbol{x}_{\text{env}}).\end{aligned} \tag{27}$$

Building upon the expression of E in (15), further simplifications are derived as follows:

$$\begin{aligned}\dot{V} = &\Delta\dot{\tilde{\boldsymbol{x}}}^\top \left(\boldsymbol{I}_6 - \varpi\left(\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon}\right)\right)\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \varpi\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ (1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top (\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon})\,\boldsymbol{\rho}_{\text{frc}}\boldsymbol{\psi}_1 \boldsymbol{F}_{\text{ctrl}} - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ (1-\varpi)\frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} + \dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right).\end{aligned} \tag{28}$$

To evaluate the stability of the energy-aware AFAC system, the following two scenarios were considered.

1) Case 1: $\Theta = \emptyset$: Under this condition, one has $\boldsymbol{\Phi} = \text{diag}\{\phi_1, \ldots, \phi_6\} = \boldsymbol{I}_6$. Thus, the relationship $(\mu\boldsymbol{I}_6 - \boldsymbol{\varepsilon}) = \boldsymbol{I}_6$ is satisfied. Consequently, (28) can be rewritten as follows:

$$\begin{aligned}\dot{V} = &(1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \varpi\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ \frac{1}{2}(1-\varpi)\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \\ &+ (1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_{\text{frc}}\boldsymbol{\psi}_1 \boldsymbol{F}_{\text{ctrl}} - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right).\end{aligned} \tag{29}$$

Using (24), we have that

$$\begin{aligned}\dot{V} \leq &(1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \varpi\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ \frac{1}{2}(1-\varpi)\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \\ &+ (1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_{\text{frc}}\boldsymbol{\psi}_1 \boldsymbol{F}_{\text{ctrl}} - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right) \\ &+ (1-\varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} - \frac{1}{2}(1-\varpi)\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \\ &- (1-\varpi)\Delta\dot{\tilde{\boldsymbol{x}}}^\top \left(\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \boldsymbol{\psi}_1 \boldsymbol{\rho}_{\text{frc}} \boldsymbol{F}_{\text{ctrl}}\right) \\ \leq &\dot{\tilde{\boldsymbol{x}}}^\top (\nu\boldsymbol{\rho}_f - \boldsymbol{I}_6)\boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} + \dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right) \\ \leq &\dot{V}_{\text{env}}\left(\boldsymbol{x}_{\text{env}}\right) < 0.\end{aligned} \tag{30}$$

Since $\Delta\dot{\tilde{\boldsymbol{x}}}^\top \left(\nu\boldsymbol{\rho}_f - \boldsymbol{I}_6\right)\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} \leq 0$ is always satisfied.

2) Case 2: $\Theta \neq \emptyset$: In this case, one has $\phi_i < 1, \forall i \in \Theta$, and $\phi_i = 1, \forall i \in \Theta_{\text{total}} \backslash \Theta$. Furthermore, it can be shown that

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$(\mu \boldsymbol{I}_6 - \boldsymbol{\varepsilon}) = \boldsymbol{\Phi}$ with $\boldsymbol{\Phi} = \text{diag}\{\phi_1, \ldots, \phi_6\}$, and thus (28) can be rewritten as follows:

$$\begin{aligned}\dot{V} = &\Delta\dot{\tilde{\boldsymbol{x}}}^\top (\boldsymbol{I}_6 - \varpi\boldsymbol{\Phi})\,\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \varpi\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ (1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\Phi}\boldsymbol{\rho}_{\text{frc}}\psi_1 \boldsymbol{F}_{\text{ctrl}} \\ &+ (1-\varpi)\frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} + \dot{V}_{\text{env}}. \end{aligned} \tag{31}$$

By incorporating (24), (31) can be modified into

$$\begin{aligned}\dot{V} \le &\Delta\dot{\tilde{\boldsymbol{x}}}^\top (\boldsymbol{I}_6 - \varpi\boldsymbol{\Phi})\,\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \varpi\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} - \Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{B}\dot{\tilde{\boldsymbol{x}}} \\ &+ (1-\varpi)\,\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\Phi}\boldsymbol{\rho}_{\text{frc}}\psi_1 \boldsymbol{F}_{\text{ctrl}} + (1-\varpi)\frac{1}{2}\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \\ &+ (1-\varpi)\nu\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\rho}_f \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} - \frac{1}{2}(1-\varpi)\Delta\tilde{\boldsymbol{x}}^\top \dot{\boldsymbol{K}}\Delta\tilde{\boldsymbol{x}} \\ &- (1-\varpi)\Delta\dot{\tilde{\boldsymbol{x}}}^\top \boldsymbol{\Phi}\left(\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \psi_1\boldsymbol{\rho}_{\text{frc}}\boldsymbol{F}_{\text{ctrl}}\right) + \dot{V}_{\text{env}} \\ \le &\Delta\dot{\tilde{\boldsymbol{x}}}^\top (\boldsymbol{I}_6 - \boldsymbol{\Phi})\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \Delta\dot{\tilde{\boldsymbol{x}}}^\top (\nu\boldsymbol{\rho}_f - \boldsymbol{I}_6)\boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} + \dot{V}_{\text{env}} \\ \le &\sum\nolimits_{i\in\Theta} (1-\phi_i)\,\rho_{f,i}\Delta\dot{\tilde{x}}_i F_{\text{err},i}. \end{aligned} \tag{32}$$

It follows that $\dot{V}$ can be decomposed into $\dot{V}_3$ and $\dot{V}_4$, such that $\dot{V} = \dot{V}_3 + \dot{V}_4$. Specifically, $\dot{V}_3$ consists of elements where $i \in \Theta$, while $\dot{V}_4$ is formed by elements where $i \in \Theta_{\text{total}}\backslash\Theta$. Moreover, $\dot{V}_4$ satisfies $\dot{V}_4 < 0$, and $\dot{V}_3$ satisfies $\dot{V}_3 \le \sum_{i\in\Theta} (1-\phi_i)\,\rho_{f,i}\Delta\dot{\tilde{x}}_i F_{\text{err},i}$. The AFAC model in (13) is shown to be passive with respect to the pair $\langle \Delta\dot{\tilde{x}}_i, F_{\text{err},i}\rangle, \forall i \in \Theta$, and asymptotic stability is guaranteed for all $i \in \Theta_{\text{total}}\backslash\Theta$. Therefore, the stability of the system is ensured.

In summary, the interconnected system composed of the AFAC model and the energy tank remains stable under various conditions. Specifically, for Case 2, as ϕ_i approaches zero, both the direct force controller and the stiffness update law are reduced to zero, thereby effectively preventing any further energy injection into the robotic system. In this limiting scenario, the inequality $|F_{e,i}| \ge F_{\text{up},i} \ge |F_{d,i}|$ holds for all $i \in \Theta_{\text{total}}$. Then, one has $\phi_i = 0, \forall i \in \Theta_{\text{total}}$, and the AFAC is transitioned to the following CAC strategy:

$$\boldsymbol{M}\Delta\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{B}\Delta\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}\Delta\tilde{\boldsymbol{x}} = \boldsymbol{F}_{\text{err}}. \tag{33}$$

Based on the inherent properties of the CAC model, when the condition $|F_{e,i}| \ge |F_{d,i}|$ is satisfied, a corresponding position update $\Delta\tilde{x}_i$ is generated, which causes $|F_{e,i}|$ to gradually decrease over time. Consequently, Case 2 progressively degenerates into Case 1, namely, the energy-aware AFAC model.

When $\Theta = \emptyset$ and $\boldsymbol{\chi} \ne \boldsymbol{\chi}^*$, it can be demonstrated that $\dot{V} < 0$, indicating that the system asymptotically converges to the equilibrium point $\boldsymbol{\chi}^*$. At the equilibrium point $\boldsymbol{\chi} = \boldsymbol{\chi}^*$, the AFAC model in (13) reduces to

$$\boldsymbol{\rho}_f \boldsymbol{F}_{\text{err}} + \boldsymbol{\rho}_{\text{frc}}\psi_1 \boldsymbol{F}_{\text{ctrl}} = \boldsymbol{0}_6. \tag{34}$$

It can be readily demonstrated that $F_{\text{err},i} = 0, \forall i \in \Theta_f$, constitutes the unique solution to the given equation. Consequently, $F_{e,i}$ is shown to converge to $F_{d,i}, \forall i \in \Theta_f$.

In free space, the conditions $\boldsymbol{\rho}_f = \boldsymbol{0}_6$, $\boldsymbol{\rho}_{\text{frc}} = \boldsymbol{0}_6$, and $\mu - \varepsilon_i = 1, \forall i \in \Theta_{\text{total}}$, hold when all directions are position-controlled. Under these conditions, $\Delta\tilde{\boldsymbol{x}} = 0$ is achieved by setting $\ddot{\boldsymbol{x}}_r(0) = \Delta\ddot{\boldsymbol{x}}_d(0)$, $\dot{\boldsymbol{x}}_r(0) = \Delta\dot{\boldsymbol{x}}_d(0)$, and $\boldsymbol{x}_r(0) = \Delta\boldsymbol{x}_d(0)$. Since $\mu - \varepsilon_i = 1, \forall i \in \Theta_{\text{total}}$, one has $\Delta\boldsymbol{x}_d = \boldsymbol{x}_d$ and $\dot{\boldsymbol{x}}_r = \dot{\boldsymbol{x}}_d$, which guarantees that the robot follows $\boldsymbol{x}_d$. Furthermore, given that industrial robots typically support only position command inputs, high-precision measurement devices, such as digital photogrammetry systems or laser trackers, can be utilized to monitor the robot's pose $\boldsymbol{x}$ in real time. The pose error can then be employed as a command for trajectory correction until the condition $||\boldsymbol{x}_d - \boldsymbol{x}|| \le \varsigma$ is satisfied. This approach guarantees the predetermined positioning accuracy in free space while maintaining the stability of the interconnected system. Thus, the proof is complete.

Remark 3 (Parameter Tuning Guidelines.): In the direction of force control, the force activation threshold $\kappa_{3,i}$ should exceed sensor noise and drift, typically 5%–20% of the desired wrench. The threshold ζ is typically set below the desired wrench tracking accuracy. The parameter δ_f controls the transition width of the activation function: a larger value yields smoother activation with better noise robustness, while a smaller value enables faster response but increases sensitivity to noise. A starting range of 1%–5% of the desired wrench is recommended. The admittance activation threshold κ_4 can be chosen as 0.5–2.0 times the positioning accuracy. The safety limit $F_{\text{up},i}$, which is defined in a task-specific manner, is chosen to be no less than the desired contact wrench to ensure compliant yet effective interaction. $V_{l,\vartheta}$ is any small positive value to avoid singularities. Additionally, as aircraft assembly is typically quasi-static and low-speed, end-effector acceleration is negligible during contact phases. In such scenarios, removing the virtual inertia term in the admittance model (i.e., setting $\boldsymbol{M} = \boldsymbol{0}$) suppresses overshoot-induced oscillations and yields smoother force regulation. Similar minimal inertia rendering strategies have been reported in [44].

The overall architecture is shown in Fig. 3. The effectiveness of the proposed method is subsequently validated through experiments involving both contact-free aircraft frame handling and contact-rich assembly tasks.

V. Experiments

To illustrate the proposed method, the experimental setup consists of a KUKA KR210 industrial robot equipped with an ATI Omega 160 F/T sensor mounted on its flange. The F/T sensor operates at a sampling frequency of 1 kHz, enabling high-resolution measurement of interaction wrenches. Although the robot is controlled by a KR C4 controller with a control cycle of 12 ms (i.e., approximately 83.3 Hz), the high-frequency force signals are filtered using a Kalman filter to suppress measurement noise, and are subsequently used in each control iteration. Additionally, a digital photogrammetric system (3-D Inspect) is employed to provide high-precision measurements of the robot's end-effector pose, as illustrated in Fig. 4. All the control algorithms for the experiments are run on an external computing platform, a laptop equipped with an Intel Core i7 processor running at 2.8 GHz and 8 GB RAM, operating in real-time.

Experiment 1 Control of contact and precise positioning—assembly: The task is to regulate the end-effector force in

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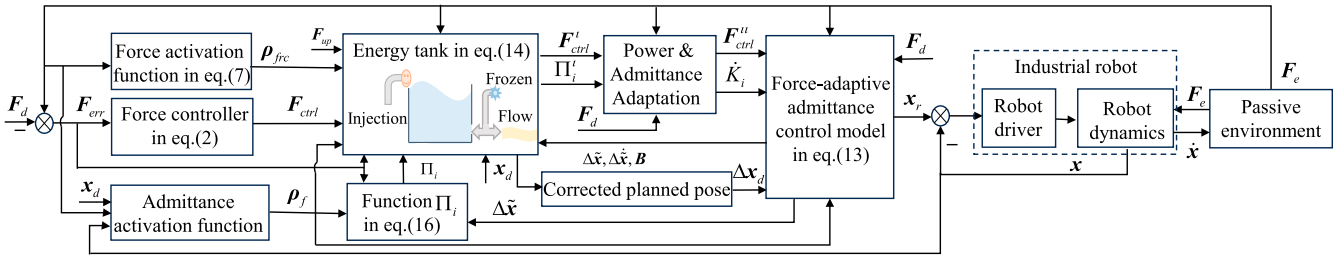


Fig. 3. Block diagram of the developed method is illustrated, where $\Pi_i^{\iota} = (\mu - \varepsilon_i)\,\Pi_i$ and $\boldsymbol{F}_{\mathrm{ctrl}}^{\iota} = (\mu \boldsymbol{I}_6 - \boldsymbol{\varepsilon})\,\boldsymbol{\rho}_{\mathrm{frc}}\boldsymbol{F}_{\mathrm{ctrl}}$ represent the corrected values of Π_i and $\boldsymbol{\rho}_{\mathrm{frc}}\boldsymbol{F}_{\mathrm{ctrl}}$, respectively, through the energy tank. Additionally, $\boldsymbol{F}_{\mathrm{ctrl}}^{\iota\iota} = (\mu \boldsymbol{I}_6 - \boldsymbol{\varepsilon})\,\boldsymbol{\rho}_{\mathrm{frc}}\boldsymbol{\psi}_1\boldsymbol{F}_{\mathrm{ctrl}}$ is obtained from $\boldsymbol{F}_{\mathrm{ctrl}}^{\iota}$ after the adaptive power flow regulation described by (17). Energy within the energy tank is either injected or frozen, as required.

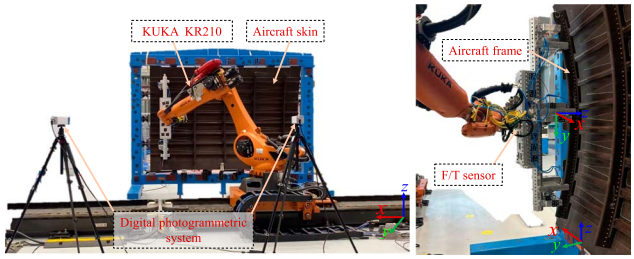


Fig. 4. Experimental setup for aircraft assembly: (a) overall schematic of the assembly scene; and (b) detailed local view of the assembly scene.

TABLE I
PERFORMANCE INDICATORS OF THE THREE ASSEMBLY SCHEMES

Metric	AFAC	Constant Energy Tank	CAC
Wrench settling time (s)	**52.31**	$> 100^{*}$	$> 100^{*}$
RMS wrench error (N)	**1.49**	9.65	10.47
Position settling time (s)	**19.41**	20.91	20.96
RMS position error (mm)	**0.04**	0.03	0.03

* Did not reach $|F_{e,2} - F_{d,2}| \leq 2\,\mathrm{N}$ and sustain it for at least 2 s within 0–100 s.
† RMS values are computed over the last 2 s.

the Y-direction while maintaining predefined positioning accuracy in the remaining directions. Specifically, the end-effector force $F_{e,2}$ is required to converge to the desired value $F_{d,2} = 100\,\mathrm{N}$, while ensuring that the position error satisfies $\|\boldsymbol{x}_p - \boldsymbol{x}_{d,p}\|_2 \leq 0.1$. Here, $\boldsymbol{x}_p = [x_1, x_3, x_4, x_5, x_6]^{\top}, \forall i \in \Theta_{\mathrm{total}} \backslash \Theta_f$, and $\boldsymbol{x}_{d,p} = [x_{d,1}, x_{d,3}, x_{d,4}, x_{d,5}, x_{d,6}]^{\top} = [3213.89, 1664.81, 179.329, -0.051, -89.28]^{\top}$ denote the actual and desired poses in the position control directions, respectively. Considering that sensor measurements are typically affected by noise, drift, and other uncertainties, $F_{e,2}$ is considered to have converged when $||F_{e,2} - F_{d,2}|| \leq 2\mathrm{N}$.

The controller parameters are set as $\boldsymbol{K}_p = 1.92\boldsymbol{I}_6$, $\boldsymbol{K}_i = 0.44\boldsymbol{I}_6$, $\boldsymbol{K}_d = 0.1\boldsymbol{I}_6$, $\zeta = 1.5$, $\kappa_1 = 0.1$, $\delta_f = 2$, $\kappa_{3,2} = 15$, $\kappa_4 = 0.1$, $\alpha_2 = 1$, $\nu = 0.01$, $V_{l,\vartheta} = 0.5\ J$, $\delta_{\mathrm{ek}} = 0.625$, $F_{\mathrm{up},2} = 130\ N$, $\eta_{4,2} = 3$, $\eta_1 = 10\,000$, $V_{u,\vartheta} = 50$, $\beta = 20$, $\lambda = 0.05$, $p_{j,i}^{+}(0) = 50$, $\boldsymbol{K}(0) = 3000$, $\boldsymbol{B} = 2\sqrt{\boldsymbol{K}}$, $\boldsymbol{M} = \boldsymbol{0}_6$, $V_{\vartheta}(0) = 2$. For, $i \in \Theta_{\mathrm{total}} \backslash \Theta_f$, $F_{d,i} = 50\,\mathrm{N}$, $\kappa_{3,i} = 45$, $\alpha_i = 0$, $\eta_{4,i} = 0.03\kappa_{3,i}$, $F_{\mathrm{up},i} = 1.3\kappa_{3,i}\,\mathrm{N}$. The initial state in the position control directions is given by $\boldsymbol{x}(0) = [3210, -1950.1, 1666.3, 179.301, -0.038, -89.271]^{\top}$.

The proposed method is compared with the AFAC approach, which employs an energy tank interconnection without energy injection and freezing functions (constant energy tank-based method), as well as the CAC method. For a fair comparison, the constant energy tank-based method is implemented with $\varepsilon_i \equiv 0$, and the CAC method is executed with $\mu = 0$ and $\varepsilon_i \equiv 0$, while all other experimental parameters remain identical.

The experimental results of the proposed method are presented in Fig. 5(a). The figure demonstrates convergence of the end-effector wrench tracking error along the Y-axis and compliance with the prescribed positioning accuracy; both metrics meet their respective specifications. As shown in Fig. 5(a), $\rho_{f,2} > 0$ at $t \approx 27.5\,\mathrm{s}$. As the contact wrench increases, both the force and admittance controllers are fully activated by $t \approx 29\,\mathrm{s}$ (i.e., $\rho_{f,2} = 1$ and $\rho_{\mathrm{frc},2} = 1$). Meanwhile, the energy tank compensates for potentially active behavior of the adaptive stiffness and force controllers, leading to a temporary decrease in stored energy. Subsequently, at $t \approx 29.5\,\mathrm{s}$, the conditions $V_{\vartheta} \leq V_{l,\vartheta} + \delta_{\mathrm{ek}}$ and $F_{e,\mathrm{sa},i} \leq -\eta_{4,i}$ are satisfied, triggering the energy-injection mechanism ε_i to replenish the tank and maintain performance; specifically, the injection enforces an effective scaling of unity, $\mu - \varepsilon_i = 1$, even when $\mu < 1$. As the wrench tracking error converges and stiffness variation diminishes, part of the energy dissipated through damping is recovered and stored in the tank, increasing its level. Finally, at $t \approx 47\,\mathrm{s}$, once $V_{\vartheta} \geq V_{l,\vartheta} + \delta_{\mathrm{ek}}$ and $F_{e,\mathrm{sa},i} \leq -\eta_{4,i}$ hold, energy injection ceases (i.e., $\mu = 1$ and $\varepsilon_i = 0$). Since ε_i, μ, $\rho_{f,i}$, and $\rho_{\mathrm{frc},i}$ are dimensionless, their numerical values characterize energy-tank and activation properties rather than physical magnitudes; accordingly, units are omitted from the plots.pt Notably, the constant energy tank baseline provides a simplified reference for stiffness adaptation. It uses the same update law as in (16) but disables energy-aware regulation by enforcing $\varepsilon_i \equiv 0$. The experimental results for this baseline are shown in Fig. 5(b). Because adaptive stiffness and direct force control can induce nonpassive behavior, the tank energy is progressively consumed to preserve overall passivity. Consequently, V_{ϑ} decreases to its lower bound $V_{l,\vartheta}$, at which point the tank modulation factor satisfies $\mu = 0$. Parameter adaptation then halts, and the controller reduces to the CAC strategy (i.e., fixed admittance). As an indirect force control approach, CAC

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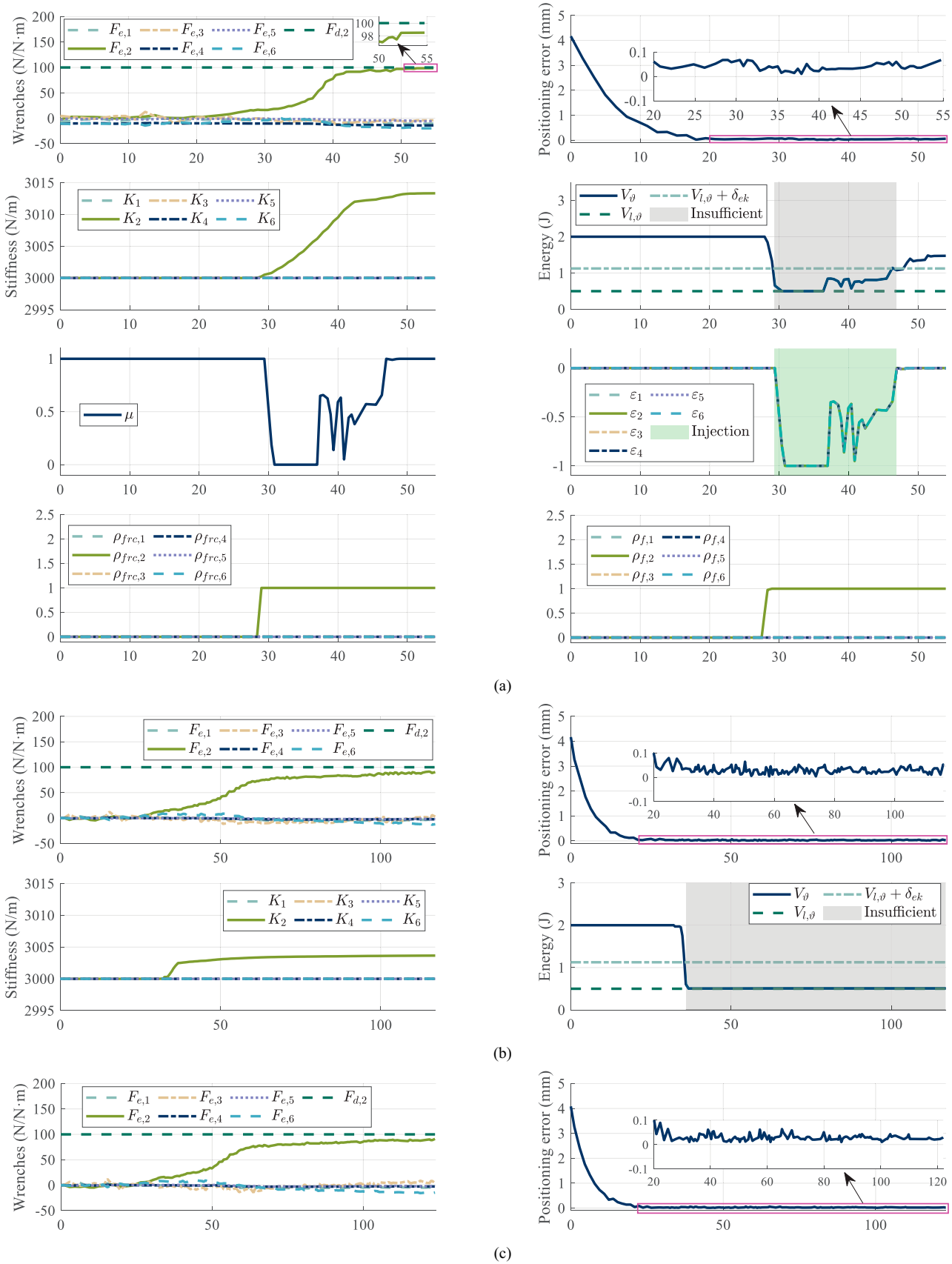


Fig. 5. Performance of the admittance device during robot-assisted aircraft frame–skin assembly under three control strategies (a)–(c). Insets in (a) show that the proposed AFAC achieves both high positioning accuracy and accurate wrench tracking, whereas the constant energy-tank-based method (b) and CAC (c) fail to maintain the assembly force within the prescribed range.

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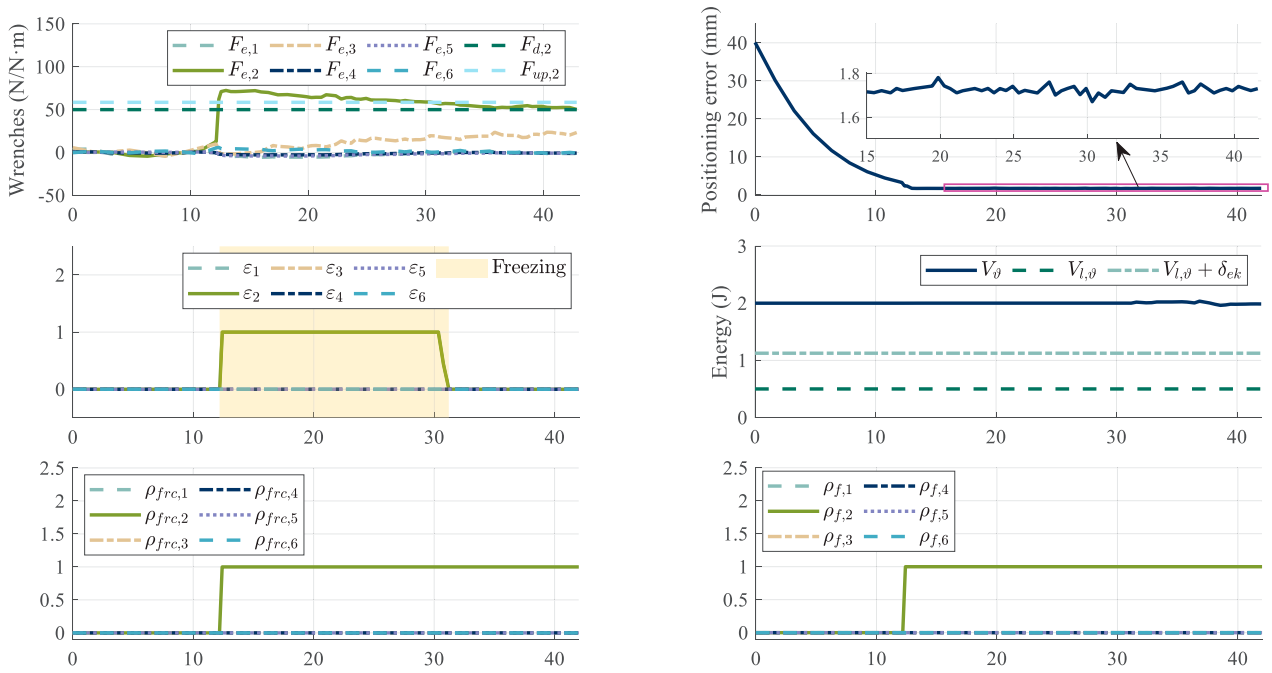


Fig. 6. System performance of the proposed method when the robot encounters unexpected contact events during free-space motion. When the contact force satisfies $F_{e,2} \geq F_{\mathrm{up},2}$, the tank's energy exchange channel (along Y) is frozen to preserve overall passivity.

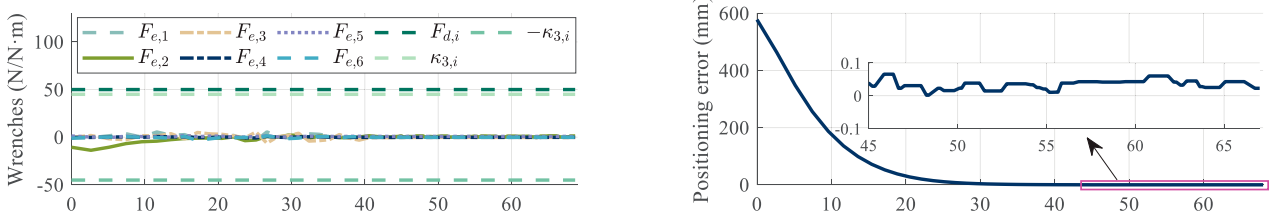


Fig. 7. System performance of the aircraft frame handling experiment. Since the contact wrench remains below the predefined threshold, the activation function outputs zero, allowing the robot to follow the planned trajectory and achieve high-precision positioning.

is highly sensitive to parameter tuning and is inherently nonrobust to environmental variations (e.g., contact stiffness and pose uncertainties). In such settings, this sensitivity often yields degraded wrench tracking performance, as evidenced by the observed wrench regulation errors. This degradation is further illustrated in Fig. 5(c), which shows the temporal evolution of wrench regulation under CAC.

Additionally, Table I lists settling time, rms wrench error, and positioning accuracy. With identical parameter settings, the proposed AFAC achieves the fastest convergence and the smallest wrench tracking error. The advantage comes from its structural design and the adaptive energy-aware mechanism, not from parameter tuning. The ablation study confirms that removing energy injection and freezing leads to a clear performance drop, underscoring their pivotal role in high-precision interaction tasks. Note that all three strategies disable corrections along the position control directions once the error falls below 0.1, ensuring a final positioning error < 0.1. Under this shared stopping rule, the proposed method may report a slightly larger residual error than the baselines; this reflects the common termination criterion rather than inferior steady-state accuracy.

Experiment 2 Robustness—compliance in the direction of position control: There are several common types of environmental uncertainties that arise in real-world robotic assembly tasks. These include contact point uncertainty caused by misalignment or surface tolerances, as well as external interaction disturbances, such as unexpected contact events occurring during free-space motion. To evaluate the robustness of the proposed method under potential environmental uncertainties, all directions were regulated under *position control*, while deliberately commanding a desired pose located *inside* an unmodelled composite skin, thus creating a severe geometric mismatch between the plan and the physical environment. The desired pose was specified as $\boldsymbol{x}_d = [3213.89,\ -2003,\ 1664.81,\ 179.329,\ -0.051,\ -89.28]^\top$, which lies within the aircraft skin. Because a noncompliant pure position controller on the 210 kg-payload KUKA KR210 would rigidly drive the end-effector toward a commanded pose that lies *inside*

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the high-stiffness aircraft skin ($>10^9\,\mathrm{N/m}$), risking hazardous impact forces and workpiece damage, only the proposed AFAC scheme was evaluated in Experiment 2. The initial pose was $\boldsymbol{x}(0) = [3210.95, -1963.03, 1666.19, 179.322, -0.008, -89.287]^\top$. The control parameters were set as $F_{d,2} = 50\,\mathrm{N}$, $\alpha_2 = 0$, $\kappa_{3,2} = 45$, $\eta_{4,2} = 0.03\kappa_{3,2} = 1.35$, and $F_{\mathrm{up},2} = 1.3\kappa_{3,2} = 58.5\,\mathrm{N}$, $\forall i \in \Theta_{\mathrm{total}}$. All other parameters were consistent with those used in Experiment 1.

The experimental results, shown in Fig. 6, illustrate the robot's approach to the aircraft skin. When contact with the skin occurs and $\Delta F_{e,2} > 0$, the activation functions $\rho_{f,2}$ and $\rho_{\mathrm{frc},2}$ become positive, prompting AFAC to issue a reference trajectory correction. At $t \approx 12.4\,\mathrm{s}$, the condition $F_{e,2} \geq F_{\mathrm{up},2}$ is satisfied, setting $\varepsilon_2 = 1$. This freezes the energy exchange channel along the Y-direction, enforcing $\mu - \varepsilon_2 = 0$. Consequently, the AFAC behavior in the Y-direction reduces to CAC, ensuring compliant contact.

At $t \approx 31\,\mathrm{s}$, as the contact wrench decreased, the condition $F_{e,2} \leq F_{\mathrm{up},2} - \eta_{4,2}$ was satisfied, prompting $\varepsilon_2 = 0$. This action reactivated the Y-direction energy exchange channel with the energy tank, and restored $\mu - \varepsilon_2 = 1$. Consequently, the AFAC in the Y-direction was reinstated, ensuring that the contact wrench converged to the desired value and thereby halting the robot's motion in the position control direction. This adjustment ensures the robot's safety. However, as a trade-off, as shown in Fig. 6, the positioning error in this experiment did not converge to the predetermined threshold.

In conclusion, the developed method effectively overcomes the limitations of offline initial task energy estimation. By continuously evaluating the system state and predefined conditions, it can adaptively determine whether to inject or freeze energy exchange in real time, thus achieving a balance between performance and safety. Moreover, it demonstrates robust performance in the presence of environmental uncertainties.

Experiment 3 Precise positioning in free space: In the third experiment, the precise positioning performance of the proposed scheme was validated in free space. The initial condition was set to $\boldsymbol{x}(0) = [3109.7, -1382.7, 1698.9, 179.318, -0.053, -84.271]^\top$, retaining the same controller parameters as in Experiment 2. The target pose for airframe handling was defined as $\boldsymbol{x}_d = [3213.89, -1950.3, 1664.81, 179.329, -0.051, -89.28]^\top$.

As shown in Fig. 7, it was observed that $|F_{e,i}| \leq \kappa_{3,i}$ for all $i \in \Theta_{\mathrm{total}}$, indicating that $\rho_{f,i} = 0$ and $\rho_{\mathrm{frc},i} = 0$. Consequently, the robot accurately followed the desired pose $\boldsymbol{x}_d$. Leveraging the measurements from the digital photogrammetric system, the positioning error $\boldsymbol{x}_d - \boldsymbol{x}$ was employed as a trajectory correction command, thereby facilitating real-time pose adjustments. Consequently, the final positioning error of the robot was maintained within the specified threshold, thus ensuring $||\boldsymbol{x}_d - \boldsymbol{x}|| \leq 0.1$. This result confirms that the activation functions effectively suppress unnecessary trajectory updates caused by measurement noise and sensor bias, thereby enhancing precision in aircraft assembly tasks.

Although Experiment 3 was conducted under nominal conditions, the proposed scheme can automatically switch to compliant behavior if unexpected contact occurs. Once the external wrench exceeds the threshold $\kappa_{3,i}$, the activation function $\rho_{\mathrm{frc},i}$ triggers the force-admittance controller, even in position-controlled directions. This behavior was confirmed in Experiment 2, where the robot safely handled an unforeseen contact with the aircraft skin.

VI. Conclusion

Simultaneous regulation of position and wrench under uncertain conditions remains a significant challenge in robotic manipulation. However, traditional hybrid force-position control approaches typically rely on precise environmental models. In this article, an energy-aware AFAC strategy is proposed to overcome this limitation, achieving simultaneous position and force regulation while ensuring safety in uncertain environments. Specifically, the AFAC model is constructed by incorporating the outputs of a force controller and an adaptive admittance module into a generalized admittance framework, thereby enabling concurrent regulation of wrench and motion to facilitate safe contact operations. Furthermore, to enhance positioning accuracy, two activation functions are introduced to preserve performance in free space.

Despite active control actions induced by stiffness adaptation and direct force controller, system passivity is maintained by employing an energy tank that monitors passivity violations. Furthermore, system performance and safety are improved through an online energy injection and freezing mechanism, which dynamically injects energy when the contact wrench is small and the tank energy is nearly depleted, and freezes energy extraction when the contact wrench exceeds a predefined threshold. Theoretical analyses are supported by experimental validation on an industrial robot for aircraft assembly. The results demonstrate that the proposed strategy not only achieves accurate position and wrench regulation but also guarantees robust safety under environmental uncertainties.

While effective, the proposed method has limitations in extreme scenarios, such as high-speed impacts or interaction with active environments, where predictive models or fast disturbance observers may be required and are beyond the present scope. Future work will extend the framework to more complex contact conditions and time-varying environments. In addition, integrating explicit disturbance-rejection schemes may further improve the robustness of the activation logic and switching behavior under sensor noise and model uncertainty.

References

[1] D. Zeng, Y. Wang, Y. Jiang, H. Tan, Z. Miao, and Y. Feng, "Distributed neural adaptive impedance control for cooperative manipulation with unknown objects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12363–12376, Jul. 2025.

[2] Y. Zhang, X. Zhao, Y. Chen, B. Tao, and H. Ding, "Observer-based variable impedance control using moving horizon estimation for robot machining thin-walled workpieces," *IEEE Trans. Ind. Electron.*, vol. 71, no. 6, pp. 5972–5982, Jun. 2024.

[3] J. Jiang, Y. Wang, Y. Jiang, H. Zhong, H. Zhang, and C. Yang, "Human-guided image feature-based visual-impedance adaptive control for intensive contact tasks," *IEEE/ASME Trans. Mechatron.*, early access, Jul. 3, 2025, doi: 10.1109/TMECH.2025.3578085.

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[4] J. Zhang, Z. Jin, Z. Zhao, and C. Yang, "A novel robotic skill learning approach for assembly task with dynamical system and broad learning," *IEEE Trans. Ind. Electron.*, vol. 72, no. 9, pp. 9304–9313, Sep. 2025.

[5] F. Ferraguti, C. Secchi, and C. Fantuzzi, "A tank-based approach to impedance control with variable stiffness," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2013, pp. 4948–4953.

[6] Y. Li, G. Ganesh, N. Jarrassé, S. Haddadin, A. Albu-Schaeffer, and E. Burdet, "Force, impedance, and trajectory learning for contact tooling and haptic identification," *IEEE Trans. Robot.*, vol. 34, no. 5, pp. 1170–1182, May 2018.

[7] N. Hogan, "Impedance control: An approach to manipulation," in *Proc. Am. Control Conf.*, 1984, pp. 304–313.

[8] J. Schimmels and M. Peshkin, "Admittance matrix design for force-guided assembly," *IEEE Trans. Robot. Automat.*, vol. 8, no. 2, pp. 213–227, Feb. 1992.

[9] K. Lu, S. Han, J. Yang, and H. Yu, "Inverse optimal adaptive tracking control of robotic manipulators driven by compliant actuators," *IEEE Trans. Ind. Electron.*, vol. 71, no. 6, pp. 6139–6149, Jun. 2024.

[10] M. Brunner, L. Giacomini, R. Siegwart, and M. Tognon, "Energy tank-based policies for robust aerial physical interaction with moving objects," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, 2022, pp. 2054–2060.

[11] Z. Zhang et al., "Hybrid force/position control of multi-mobile manipulators for cooperative operation without force measurements," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol. 71, no. 1, pp. 397–410, Jan. 2024.

[12] S. Ding, J. Peng, J. Xin, H. Zhang, and Y. Wang, "Task-oriented adaptive position/force control for robotic systems under hybrid constraints," *IEEE Trans. Ind. Electron.*, vol. 71, no. 10, pp. 12612–12622, Oct. 2024.

[13] M. Iskandar, C. Ott, A. Albu-Schäffer, B. Siciliano, and A. Dietrich, "Hybrid force-impedance control for fast end-effector motions," *IEEE Robot. Automat. Lett.*, vol. 8, no. 7, pp. 3931–3938, Jul. 2023.

[14] M. H. Raibert and J. J. Craig, "Hybrid position/force control of manipulators," *J. Dyn. Syst. Meas. Contr.*, vol. 103, no. 2, pp. 126–133, 1981.

[15] S. Haddadin and E. Shahriari, "Unified force-impedance control," *Int. J. Robot. Res.*, vol. 43, no. 13, pp. 2112–2141, 2024.

[16] M. T. Mason, "Compliance and force control for computer controlled manipulators," *IEEE Trans. Syst., Man, Cybern.*, vol. 11, no. 6, pp. 418–432, Jun. 1981.

[17] Y. Lin, Z. Chen, and B. Yao, "Unified motion/force/impedance control for manipulators in unknown contact environments based on robust model-reaching approach," *IEEE/ASME Trans. Mechatron.*, vol. 26, no. 4, pp. 1905–1913, Apr. 2021.

[18] H. West and H. Asada, "A method for the design of hybrid position/force controllers for manipulators constrained by contact with the environment," in *Proc. IEEE Int. Conf. Robot. Autom.*, vol. 2, 1985, pp. 251–259.

[19] D. Zeng, Z. Liu, C. L. P. Chen, Y. Wang, Y. Zhang, and Z. Wu, "Adaptive neural prescribed-time control of switched nonlinear systems with mode-dependent average dwell time," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 53, no. 12, pp. 7427–7440, Dec. 2023.

[20] V. Ortenzi, R. Stolkin, J. Kuo, and M. Mistry, "Hybrid motion/force control: A review," *Adv. Rob.*, vol. 31, no. 19–20, pp. 1102–1113, 2017.

[21] Y. Karayiannidis and Z. Doulgeri, "Adaptive control of robot contact tasks with on-line learning of planar surfaces," *Automatica*, vol. 45, no. 10, pp. 2374–2382, 2009.

[22] J. Hathaway, A. Rastegarpanah, and R. Stolkin, "Learning robotic milling strategies based on passive variable operational space interaction control," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3435–3448, Mar. 2024.

[23] D. Ye, C. Yang, Y. Jiang, and H. Zhang, "Hybrid impedance and admittance control for optimal robot–environment interaction," *Robotica*, vol. 42, no. 2, pp. 510–535, 2024.

[24] Y. Li, L. Zheng, Y. Wang, E. Dong, and S. Zhang, "Impedance learning-based adaptive force tracking for robot on unknown terrains," *IEEE Trans. Robot.*, vol. 41, pp. 1404–1420, 2025.

[25] X. Xing, E. Burdet, W. Si, C. Yang, and Y. Li, "Impedance learning for human-guided robots in contact with unknown environments," *IEEE Trans. Robot.*, vol. 39, no. 5, pp. 3705–3721, May 2023.

[26] Z. Jin, D. Qin, A. Liu, W-a Zhang, and L. Yu, "Model predictive variable impedance control of manipulators for adaptive precision-compliance tradeoff," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 2, pp. 1174–1186, Feb. 2023.

[27] J. Wachter, R. Mikut, and F. Buse, "Modeling and force control of a terramechanical wheel-soil contact for a robotic manipulator used in the planetary rover design process," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2019, pp. 560–565,

[28] T. Sun, J. Sun, B. Lian, and Q. Li, "Sensorless admittance control of 6-DOF parallel robot in human-robot collaborative assembly," *Rob. Comput. Integr. Manuf.*, vol. 88, 2024, Art. no. 102742.

[29] D. Lee, D. Ko, M. J. Kim, and W. K. Chung, "Bidirectional energy flow modulation for passive admittance control," *IEEE Trans. Robot.*, vol. 40, pp. 3917–3932, 2024.

[30] C. Secchi, S. Stramigioli, and C. Fantuzzi, "Position drift compensation in port-Hamiltonian based telemanipulation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2006, pp. 4211–4216,

[31] J. Fu, I. Burzo, E. Iovene, J. Zhao, G. Ferrigno, and E. De Momi, "Optimization-based variable impedance control of robotic manipulator for medical contact tasks," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–8, 2024.

[32] H. Zhou, X. Zhang, J. Liu, and Z. Ju, "A stability-guaranteed variable admittance control architecture for complex physical interaction tasks with multiple scenarios," *IEEE Trans. Syst., Man, Cybern.: Syst.*, vol. 55, no. 3, pp. 1746–1758, Mar. 2025.

[33] Y. Michel, M. Saveriano, and D. Lee, "A novel safety-aware energy tank formulation based on control barrier functions," *IEEE Robot. Automat. Lett.*, vol. 9, no. 6, pp. 5206–5213, Jun. 2024.

[34] B. Capelli, C. Secchi, and L. Sabattini, "Passivity and control barrier functions: Optimizing the use of energy," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 1356–1363, Feb. 2022.

[35] F. Benzi, F. Ferraguti, G. Riggio, and C. Secchi, "An energy-based control architecture for shared autonomy," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3917–3935, Jun. 2022.

[36] Y. Huang and L. Yang, "Power adaptation-enabled admittance control for stable and safe actuated interaction in unmodeled environment," *IEEE Robot. Automat. Lett.*, vol. 10, no. 3, pp. 2798–2805, Mar. 2025.

[37] C. Schindlbeck and S. Haddadin, "Unified passivity-based cartesian force/impedance control for rigid and flexible joint robots via task-energy tanks," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2015, pp. 440–447,

[38] F. Benzi, M. Brunner, M. Tognon, C. Secchi, and R. Siegwart, "Adaptive tank-based control for aerial physical interaction with uncertain dynamic environments using energy-task estimation," *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 9129–9136, Apr. 2022.

[39] R. Rashad et al., "Energy aware impedance control of a flying end-effector in the port-Hamiltonian framework," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3936–3955, Jun. 2022.

[40] K. Bai, W. Chen, K.-M. Lee, Z. Que, and R. Huang, "Spherical wrist with hybrid motion-impedance control for enhanced robotic manipulations," *IEEE Trans. Robot.*, vol. 38, no. 2, pp. 1174–1185, Feb. 2022.

[41] D. Zeng, Z. Liu, Y. Wang, C. L. P. Chen, Y. Zhang, and Z. Wu, "Adaptive fuzzy inverse optimal control of nonlinear switched systems," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 9, pp. 3093–3107, Sep. 2023.

[42] E. Shahriari, L. Johannsmeier, E. Jensen, and S. Haddadin, "Power flow regulation, adaptation, and learning for intrinsically robust virtual energy tanks," *IEEE Robot. Automat. Lett.*, vol. 5, no. 1, pp. 211–218, Jan. 2020.

[43] P. B. G. Dohmann and S. Hirche, "Distributed control for cooperative manipulation with event-triggered communication," *IEEE Trans. Robot.*, vol. 36, no. 4, pp. 1038–1052, Apr. 2020.

[44] A. Q. Keemink, H. Van der Kooij, and A. H. Stienen, "Admittance control for physical human–robot interaction," *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1421–1444, Nov. 2018.

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